Metaraminol

Mechanism of action

- Metaraminol is an alpha receptor agonist. It stimulates alpha-1 receptors in peripheral blood vessels, causing vasoconstriction and a rise in blood pressure.
- Some references describe a small degree of stimulation of beta receptors, but this is usually clinically insignificant.

Authorised practice levels

- ICPs and CCPs.

Indications

- ✓ Hypotension in the setting of septic shock, post cardiac arrest, cardiogenic shock, adrenal crisis, severe traumatic brain injury, neurogenic shock, rapid sequence intubation and post intubation.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ Bradycardia. Metaraminol may make bradycardia worse as reduced endogenous release of adrenaline may occur as blood pressure is raised. Consider administering adrenaline if the patient is hypotensive and bradycardic.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated. The likelihood of a pregnant or breastfeeding patient requiring metaraminol is very low, but it should be administered if indicated.

Dosage

- Titrate the dose and frequency to the patient's blood pressure:
  - » Administer an infusion via an infusion pump, or
  - » Administer 0.5-1 mg IV every 5-10 minutes as required for an adult, or

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- See the paediatric drug dose tables for bolus doses for a child, and administer IV every 5-10 minutes as required.

- If an infusion is administered in an adult:
  - Administer an initial bolus of 0.5-1 mg IV.
  - Start the infusion at 2 mg/hour and adjust the rate as required.
  - The patient is likely to require approximately 2-3 mg/hour.

Administration

- IV bolus administration in an adult:
  - Dilute 10 mg of metaraminol to a total of 10 ml using 0.9% sodium chloride. This final solution contains 1 mg/ml.
  - Administer the required dose, preferably via a running line.

- IV bolus administration in a child:
  - Draw up 1 ml of metaraminol from a 10 mg/1 ml ampoule.
  - Place into a 100 ml bag of 5% glucose.
  - Shake well and label. This final solution contains 0.1 mg/ml.
  - See the paediatric drug dose tables for dosing/volume and draw up doses from the bag of 5% glucose.

- IV infusion:
  - Follow service guidelines for dilution and infusion pump set up.
  - Adjust the infusion rate as required.

Common adverse effects

- Hypertension. This usually only occurs when boluses greater than 1 mg are administered in an adult.

- Bradycardia. See cautions.

Usual onset of effect

- 1-2 minutes.

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Usual duration of effect

- 10-15 minutes.

Usual preparation

- Ampoule containing 10 mg in 1 ml.
- Pre-filled syringe containing 10 mg in 10 ml.

Pharmacokinetics

- Metaraminol is predominantly metabolised in the liver with metabolites being excreted in the urine.
- Liver impairment does not alter the dose which is titrated to effect.

Common interactions

- None.

Additional information

- Some references describe the IM administration of metaraminol but this has no role in the out-of-hospital setting.

More Information

Version history

9 June 2025 1.0.5.3
Update: Indications updated.

22 November 2022 1.0.1
Update: Usual preparation and administration.